Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI engineer with 6+ years of production systems experience, now building LLM guardrails and evaluation harnesses at Intuit. Built AgentProbe -- a 15-endpoint LLM evaluation toolkit scoring factual accuracy, hallucination rate, retrieval quality, completeness, and compliance via LLM-as-judge across curated labeled datasets (1,240+ test cases with train/test/validate splits). Built LeaseI with a guardrail layer enforcing that LLM never auto-resolves Fair Housing or eligibility decisions -- safety boundary validated via adversarial testing (93.6% precision, 87.9% F1). Spent 8 years in **Fair Housing Act-compliant affordable housing systems at Entrata** (LIHTC, HUD, HAP, Housing Choice Voucher) -- the exact domain RealAssist operates in. Case study: github.com/venkatalakshmiparimala/AgentProbe. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

LLM-as-Judge Evaluation | Guardrail Systems | AI Safety Layer Design | Fair Housing Act Compliance | Precision/Recall/F1 | Labeled Eval Datasets | Prompt Injection Defense | Red-teaming | CI Safety Regression | Terraform / IaC | Python (9+ yrs) | PropTech Domain

EXPERIENCE

Intuit

Jun 2026 - Present

Software Engineer · Virginia, USA

- Build **LLM guardrails** for Anthropic Claude in production financial apps -- prompt injection filtering, input validation, sensitive-data handling, and fallbacks when model output fails safety checks.
- Design and run **LLM-as-judge evaluation harnesses** with golden-dataset CI checks and regression tests to catch hallucinations and safety regressions before release; track precision/recall per failure category.
- Integrate Anthropic Claude into Python/FastAPI services for financial data summaries and natural-language queries; design structured-output schemas so answers are traceable to source records.
- Develop backend APIs and data pipelines with React/TypeScript front ends deployed to AWS (S3, EC2, Lambda, API Gateway) via GitHub Actions CI/CD.
- Build ML pipelines (scikit-learn, XGBoost) covering data ingestion, feature engineering, model evaluation, and prediction APIs.

Entrata

Oct 2020 - Jul 2024

Senior Software Engineer · Multi-tenant enterprise SaaS, 100K+ monthly active users

- Owned **Fair Housing Act-compliant affordable housing systems for 8 years** (LIHTC, HUD income limits, HAP, Housing Choice Voucher, Tax Credit) -- the exact compliance domain governing RealAssist; zero violations across 7 annual federal import cycles.
- Maintained the yearly HUD income-limit import for 7 years -- loading federal data into ~180,000 domain records with validation, reconciliation, and rollback; any import error cascades to incorrect eligibility decisions for 100K+ housing units.
- Delivered 20+ features end-to-end (Python, PHP, Java, JavaScript, React, PostgreSQL) across 120+ Agile sprints, including Python/Bash automation that eliminated 80% of manual operational workload.
- Optimized 200+ SQL queries using execution plans and indexing (30% performance gain); AWS S3 migration with Libsodium encryption for 100% PHI compliance, zero security incidents.
- Docker microservices redesign (50% better scalability, 30% faster deployments); reviewed 300+ pull requests and mentored junior engineers.

Entrata India

Sep 2018 - Oct 2020

Junior Software Engineer · Pune, India

- Built property management modules (Python, PHP, Java, PostgreSQL) for affordable housing, rental management, and large-scale data reconciliation (1.6M HUD/Tax Credit records).
- Led AWS S3 cloud migration from legacy server storage; created 56 state-specific Fair Housing certification documents.

PROJECTS

AgentProbe -- Open-Source LLM Evaluation Toolkit

Live on GitHub

Built a homegrown **guardrail evaluation harness**: 15-endpoint FastAPI backend routing prompts across five LLM providers (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama) with a 5-metric LLM-as-judge scoring factual accuracy, hallucination rate, retrieval quality, completeness, and compliance. Curated labeled eval datasets with train/test/validate splits; CI regression checks (GitHub Actions) alert when metrics degrade. Used to evaluate Leasel across 1,240+ test cases.

Python · FastAPI · LangGraph · ChromaDB · LangChain · Anthropic · OpenAI · Azure OpenAI · Groq · GitHub Actions -- github.com/venkatalakshmiparimala/AgentProbe

Leasel -- Multi-Agent AI Platform with Safety Guardrails

Built a **compliance AI safety layer** for affordable housing: guardrails enforced in code so the LLM never auto-resolves Fair Housing eligibility, rent calculations, or compliance decisions -- safety boundary validated via **adversarial (red-team) test cases**. LLM-as-judge CI harness: **93.6% precision, 87.9% F1 across 1,240 test cases** with train/test/validate splits and confusion-matrix analysis per failure category. 8-agent LangGraph workflow (triage, retrieval, reasoning, validation) with HITL escalation; hybrid-search RAG (TF-IDF + semantic) with cited answers.

Python · LangGraph (8-agent) · FastAPI · Streamlit · TF-IDF RAG · OCR · SQLite · CI/CD -- github.com/venkatalakshmiparimala/Lease_complaince_rag

AI on FHIR -- Clinical NLP with HIPAA-Compliant Guardrails

Built LLM guardrails for clinical data: role-based access control (HIPAA-style data isolation) ensuring patient data is only accessible to authorized roles, input validation, and adversarial testing against privilege-escalation attempts. Translates English/Spanish questions into FHIR API calls via LLMs and BERT, cutting manual lookup time by ~60%.

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs -- github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — May 2026

GPA 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2017

TECHNICAL SKILLS

AI Safety & Evaluation: LLM-as-Judge, Guardrail Evaluation Pipelines, Prompt Injection Defense, Jailbreak Defense, OWASP LLM Top 10, Adversarial Testing, Red-teaming, Content Moderation, AI Safety Layer Design, Overblocking Evals, CI Safety Regression

LLMs & Agents: Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama, RAG Pipelines, Hybrid Search, Embeddings, Prompt Engineering, LangGraph, LangChain, HITL, MCP

Evaluation Frameworks: Homegrown harness (AgentProbe), DeepEval-compatible patterns, RAGAS-compatible patterns, labeled dataset curation, train/test/validate splits, confusion matrix analysis, precision/recall/F1

Vector Search & NLP: ChromaDB, Pinecone, Weaviate, Elasticsearch, Semantic Search, BERT, spaCy, HuggingFace

Python & ML: Python (9+ yrs), FastAPI, Flask, Django, Pandas, NumPy, scikit-learn, XGBoost, PyTorch

Web & Other Languages: TypeScript, JavaScript, React, Node.js, Java, Spring Boot, PHP, SQL, PL/SQL

Databases & Data: PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, Snowflake, ETL Pipelines, Query Optimization

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, AKS), GCP (Vertex AI), Docker, Kubernetes, **Terraform**, GitHub Actions CI/CD

Compliance & Security: Fair Housing Act, LIHTC, HUD, HAP, HIPAA/PHI, OAuth, Secure Coding, Observability, Monitoring